

SIMULATION PIPELINE & MATHEMATICAL ARCHITECTURE

From Capacitor Discharge Inputs to Receptance Coupling & Chatter Stability Outputs

STAGE 1: PHYSICAL & EXPERIMENTAL INPUT PARAMETERS

Coilgun Inputs

- Charging Voltage: V in $[20, 80]$ V
- Capacitance: $C = 470$ uF
- Efficiency: $\eta = 8.2\%$
- Striker: $m_p = 0.883$ g, $R = 3$ mm

Electrical parameters

Target & Interface Inputs

- Steel Target: $E = 210$ GPa, $\nu = 0.3$
- PU Pad (Shore A90): $E = 42$ MPa
- Pad thickness = 5 mm
- Sampling: $f_s = 20$ MHz ($dt = 50$ ns)

Contact interface

Substructure Inputs

- Spindle Bearing Dynamics G_s
- Module Free-Free FRF Matrix: H_{ij}
- Cutting Tool Geometry (CAD)
- Milling Coeffs: $K_s = 800$ MPa, $N_t = 2$

Structural matrices

STAGE 2: COMPUTATIONAL SIMULATION ENGINES (NUMERICAL SOLVERS)

Sim 1: Electromechanical ODE

1. Electrical Stored Energy:
 $E_0 = 0.5 * C * V^2$
2. Striker Impact Velocity:
 $v_0 = V * \sqrt{\eta * C / m_p} \sim 0.075 * V$
3. Hertz Peak Force Scaling:
 $F_{max} = (5/4)^{(3/5)} * kH^{(2/5)} * m^{(3/5)} * v_0^{(6/5)}$

v_0, F_{max}

Sim 2: Hertz Contact & FFT

1. Non-linear Impact ODE Solver:
 $m_{eff} * x''(t) + k_H * x(t)^{1.5} = 0$
2. FFT Spectrum Analysis (20 MHz):
 $F(f) = \text{integral } F(t) * \exp(-j2\pi f t) dt$
3. Mechanical Low-Pass Filtering:
Attenuates 3.8 kHz (-20.8 dB) & 7.2 kHz

Force Spectrum $F(f)$

Sim 3: Inverse IRC & Stability

1. Inverse Receptance Coupling (IRC):
 $G_s = [H_{12}^{-1} * (H_{11} - G_{mm}) * H_{21}^{-1}]^{-1} - H_{22}$
2. Forward Tool Receptance Coupling:
 $G_{11} = \text{Substructure_Coupling}(G_s, \text{Tool})$
3. Altintas-Budak Chatter Stability Lobes

Outputs Fig 3

Outputs Fig 4

Outputs Fig 5

STAGE 3: VERIFIED EXPERIMENTAL OUTPUTS & REPRODUCED FIGURES

Figure 3: Coilgun Characterization

- Subplot (a): Bench Setup Flow
- Subplot (b): v_0 (1.5 to 6 m/s) & F_{max} (12 to 48 N) vs Voltage (20-80V)
- Subplot (c): Force Profiles:
 - Bare steel: $\tau_c = 14$ us, $F_{max} = 298$ N
 - Soft pad: $\tau_c = 400$ us, $F_{max} = 50$ N
 - Hammer metal: 250 us / Plastic: 600 us
- Subplot (d): Spectrum & Bandwidth:
 - Mode 3.8 kHz attenuated by -20.8 dB
 - Mode 7.2 kHz attenuated by -40.8 dB

Figure 4: 100-Shot Repeatability

- Subplot (a): Manual Hammer
 - High operator scatter & hand bias
 - $\text{std}(F_{max}) = 7.334$ N
 - $\text{std}(\tau_c) = 0.019$ ms (19 us)
- Subplot (b): Gauss Exciter
 - Precise electronic trigger
 - $\text{std}(F_{max}) = 1.258$ N (82.8% reduction!)
 - $\text{std}(\tau_c) = 0.002$ ms (89.5% reduction!)
 - Rapid H1 SNR convergence $\sim \sqrt{N}$

Figure 5: Tool FRF & Chatter SLD

- Subplot (a): Tool-Tip FRF Synthesis
 - Reference direct hammer: $G_{11,ref}$
 - Raw automated: exhibits parasitic mode
 - IRC-corrected: eliminates module mode, recovers true spindle dynamics (< 2% err)
- Subplot (b): Chatter Stability Lobe
 - Speed RPM vs Axial Depth a_{lim} (mm)
 - Validated with experimental tests:
 - * Stable points (Circles O)
 - * Chatter points (Crosses X)

KEY SCIENTIFIC TAKEAWAY & ENGINEERING VALUE

1. Deterministic Bandwidth: Polyurethane pad acts as a mechanical filter, shaping pulse to < 3 kHz and suppressing parasitic modes.
2. Superior Repeatability: 83-90% reduction in force/duration scatter enables automated unattended in-situ spindle tap testing.
3. Decoupling Fidelity: Inverse Receptance Coupling (IRC) completely removes measurement module dynamics from identified spindle FRF.